

Advanced Methods in Text Analytics

Word Representations



What are representations? (1)

- Let's **represent** a hypothetical country called Countryland!
- What **features** should we use?
 - Area size? E.g. 400.000 km²
 - Population size? E.g. 20M
 - Continent? E.g. Europe
- Let's group those features as a single 3D **feature vector**
 - Countryland: (400.000; 20M; 'Central Europe')
- Features can have different types
 - Real or integer numbers, e.g. population size
 - Categorical, e.g. the continents
- We refer to that vector as a **representation** of 'Countryland'
- Is that useful for, say, inferring size of the economy?

What are representations? (2)

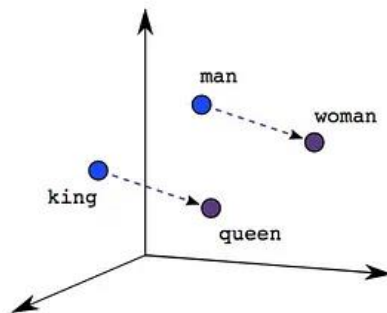
- How about the following features then?
 - GDP? E.g. 50 billion USD
 - Interest rate? E.g. 2.2%
 - Inflation rate? E.g. 7.2%
- This gives us a different representation of Countryland:
 - (50B; 2.2; 7.2)
- Manually constructing representations: **feature engineering**
 - Often requires expertise in the area
 - E.g. 'government debt to GDP' is a common metric in Economics
- **Feature vectors:** most typical form of input to machine learning (ML) methods

Can we *learn* useful representations?

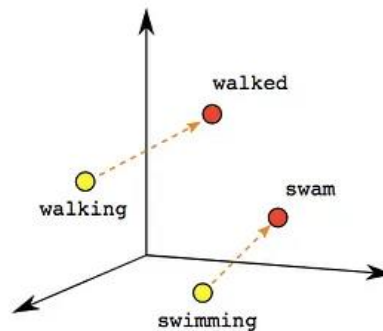
- Can we **learn useful features**?
 - Then no more feature engineering
 - We may not even need an Economics degree to get useful features
- **Yes!** Deep Learning (DL) methods are useful for this.
 - But ML/DL is mostly engineering, so back to square one?
 - No! These methods are applicable to any domain, e.g. economics, applied mathematics, etc.
- The **learned representations** may be *generally* useful, e.g. word representations
 - Potentially useful for any application with text as input

Word Embeddings

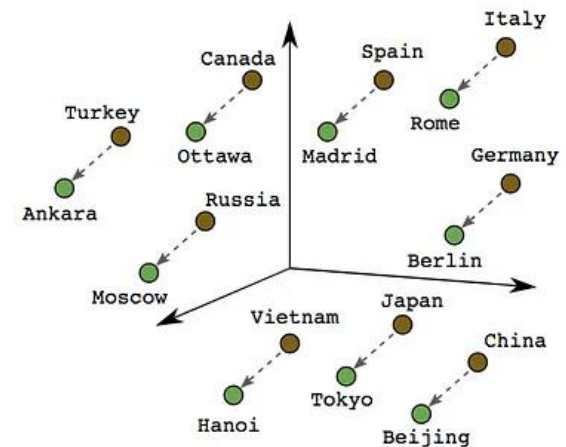
- Learned vector representations of words/tokens
- Can be learned in several different ways
- Classic example: **word2vec**
 - Useful geometrical properties between related word vectors



Male-Female



Verb Tense



Country-Capital

[Image source](#)

Challenges

- Not straightforward as previous examples suggest!
 - Vector spaces with hundreds or thousands of dimensions
 - Variety in methods for learning word representations
- Thus, many challenges!
 - Here's some examples of those challenges
- **Interpretability:**
 - Learned features often not interpretable (real numbers)
 - Geometric properties of space not straightforward (high-dimensional space)
- **Cost:**
 - Train on terabytes of text
 - Latest LLMs have *trillions* of parameters
 - GPU memory limited
- This **fixed (static) word -> vector mapping seldom used** anymore
 - But still part of architectures, relevant to understand representations

Outline

1. Machine Learning Refresher
2. Deep Learning Basics
3. Word Embeddings

Machine Learning Refresher

About this Section

- **Goal:** discuss basic concepts and terms that will be useful for the entire course
- **Outline:**
 1. Basic Machine Learning Concepts
 2. Probability Theory Refresher
 3. Maximum Likelihood Estimation
 4. Summary

What is Machine Learning? (1)

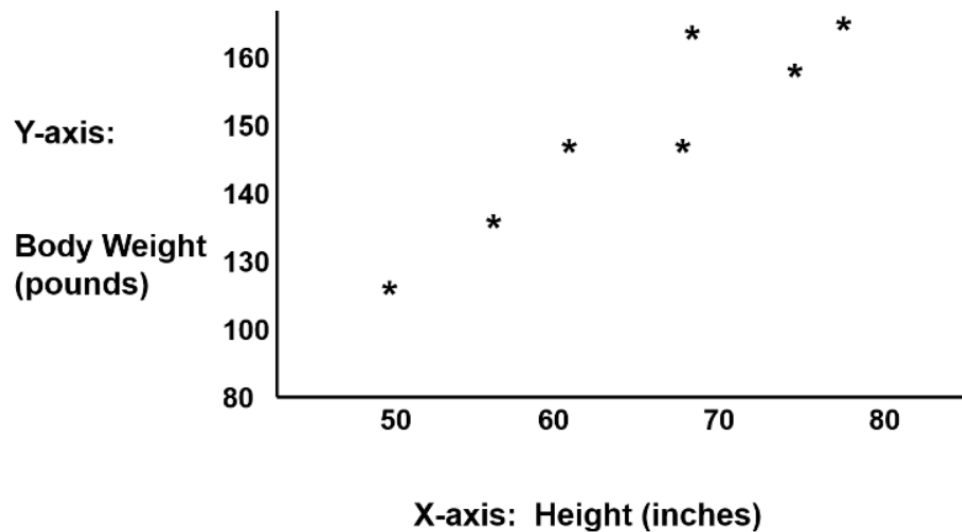
- Methods for developing algorithms by **learning from data**
 - In contrast to manually coding knowledge we already have
 - Example: flying a helicopter (how many situations must we consider?)
- Umbrella term encompassing **many families of methods**
 - E.g. Inductive Logic Programming, Deep Learning, etc.
- Related (umbrella) terms
 - **Artificial Intelligence:** solutions that show intelligent behavior
 - **Data Mining:** extract value from data (ML usually part of that pipeline)
- Machine learning **in this course?**
 - Mostly **deep learning and related methods**

What is Machine Learning? (2)

- Classic definition:
 - “A computer program is said to learn from **experience** E with respect to some class of **tasks** T and **performance measure** P , if its performance, as measured by P , improves with experience E .”
- Tom Mitchell, Machine Learning, 1997
- **Experience:** data in some format
 - In our case, text (that needs to be processed so models can "read" it)
 - **Tasks:** depends on perspective
 - ML: classification, regression, etc.
 - NLP: machine translation, question answering, etc.
 - **Performance measure:** depends on task
 - E.g. F1 score for classification tasks, ROUGE scores for summarization

What is a Task?

- **Task:** predict weight of a person (regression, supervised)
- **Data:** How do we represent a person?
 - Let's go with a simple 2D vector: (height, weight)
- We might see a linear relation between input and target
 - **Input:** 1D feature vector (usually denoted by $\mathbf{x}_i$, here: height)
 - **Target:** weight (usually denoted by $\mathbf{y}$, here: real number)



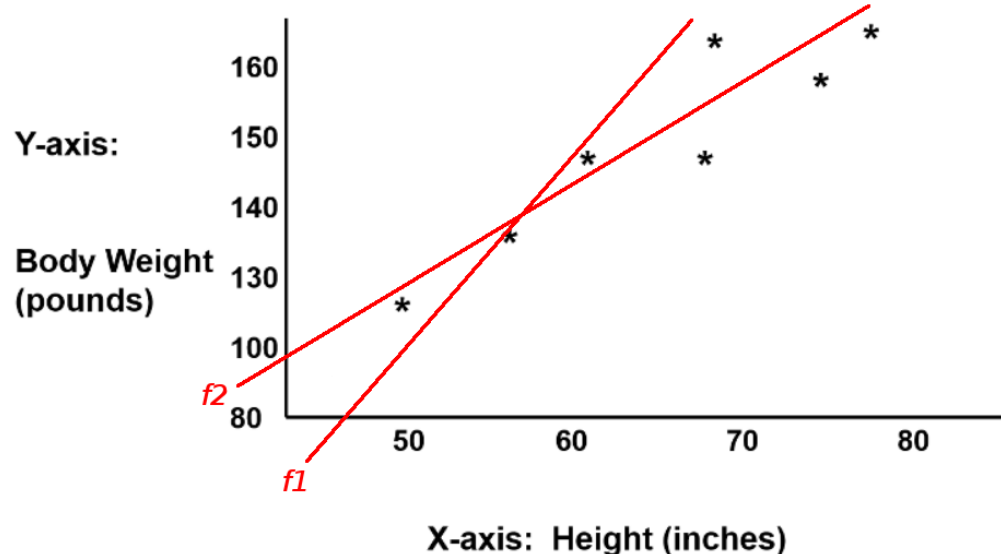
[Example source](#)

What is a Model?

- Family of methods that **we choose**
 - Each comes with a set of **assumptions**
 - E.g. based on observed data, assume linear relation between height and weight
 - Thus, we choose a linear regression model
- **Linear regression:**
 - **Main assumption:** linear relation between inputs and output
 - **Formally:** learn function $f(x) = y$
 - Here: $f(\text{height}) = \text{weight}$
 - This model comes from statistics, studied more formally, e.g. more assumptions, ML more "relaxed", focuses on learning (generalizing)
- How exactly do we **learn** a function?
 - To get there, we need to talk about parameters

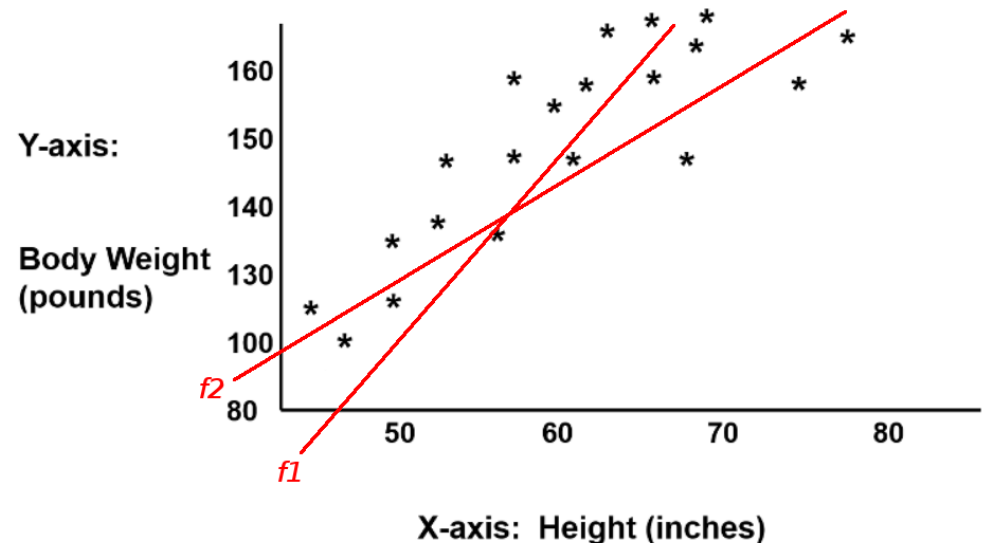
What are Parameters?

- We can describe assumption in linear regression with **equation of a line**:
 - $y = mx + b$ where m is slope and b is y -intercept
 - In our case: $weight = m \cdot height + b$
 - Thus, we have **two parameters: m, b**
- Each value assignment of m, b : different member of linear regression family
 - Which one **fits** the data better?
 - f_1 or f_2 ?
 - How do you know? Depends on the goal



What is Generalization?

- **Goal of ML:** model should perform well at the task on new data
 - Not just data it was trained on
 - Thus, goal of training a model: **generalize to unseen data**
- For example, which function did you say fits our data best?
 - f_1 or f_2 ?
- What if we collect more data?
 - Before, f_2 seemed best
 - With more data, f_1 is best
- How do we train for generalization?
 - We **evaluate models on *held-out evaluation data***, represents unseen data
 - Train and eval data assumed from same distribution



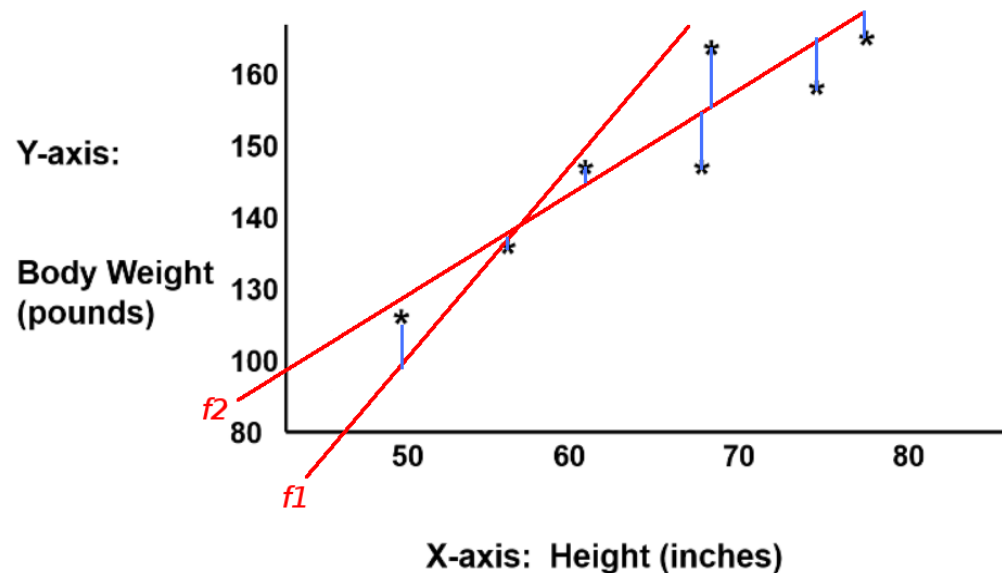
How do we Evaluate a Model?

- How can we tell which member of the family is more suitable for our task?
 - We need to **select a model** from the many possible family members
- **Performance metric**
 - A **metric we choose to evaluate** how good a **model** is at a task
 - **For example:** mean squared error (MSE)

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n \left(Y_i - \hat{Y}_i \right)^2$$

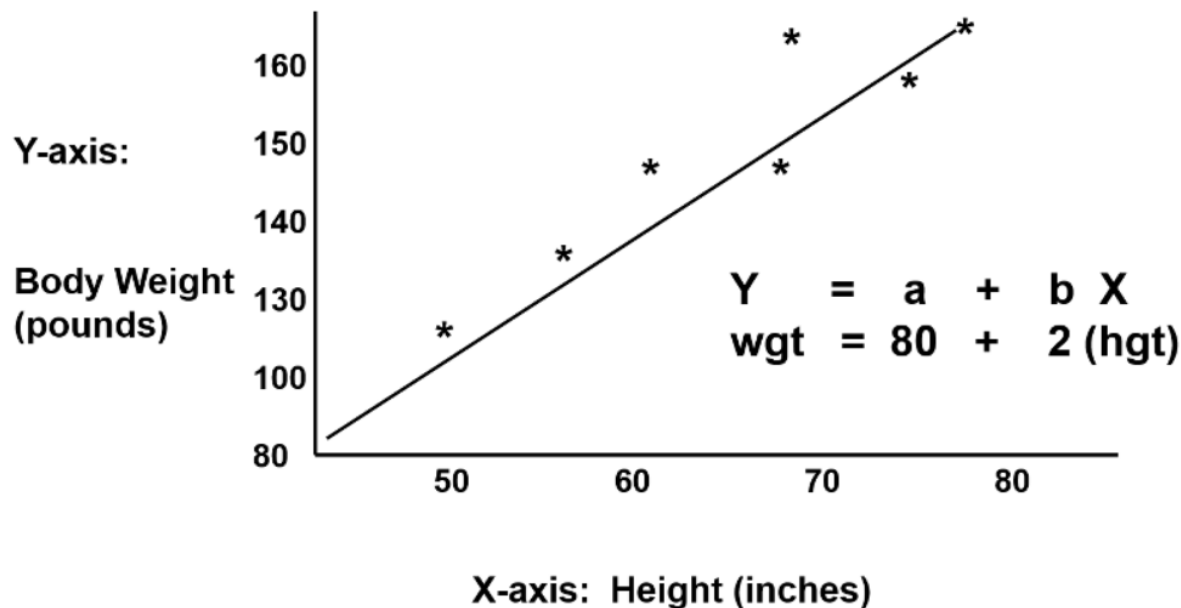
where:

- n : # of examples
- Y_i : target values
- Y'_i : predicted values



How do we Use our Selected Model?

- **Model selection:** choosing a value assignment for our parameters
 - E.g. $m = 80$, $b = 2$
- To make a **prediction**, we make use of our selected model (function)



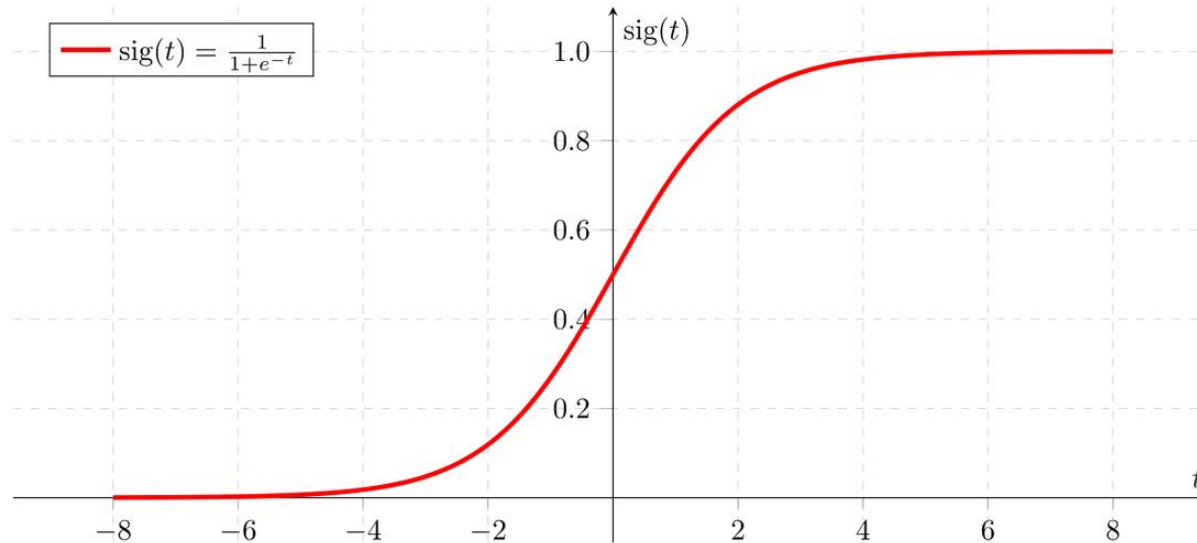
- Important distinction: inference vs prediction
 - **Inference:** use model to learn about data generation process
 - **Prediction:** predict specific outcomes for new input points

What are *Hyper*parameters?

- Other parameters that are **typically not learned**
 - Instead, they are manually set by the user
 - Note that *meta-learning* does exist, e.g. [AutoML](#)
- Example in linear regression: using a bias or not
 - No bias: line passes by origin (assumes data is centered)
 - See scikit-learn's [documentation](#) on linear regression (`fit_intercept`)
- How do we choose values for hyperparameters?
 - **Hyperparameter optimization methods**
 - E.g. grid search, random search ([known](#) to be better than grid search)
 - Requires familiarity with known tricks/experience/expert knowledge
- Often **involves making assumptions**
 - E.g. this setting worked well in the past
 - E.g. the search space I designed benefits all models equally

What about Classification?

- **Logistic regression:** popular model for *classification*
 - **Main assumption:** linear relation between inputs and output
- Essentially: linear regression with a logistic function
 - $y = \sigma(m x + b)$ where σ is the logistic (sigmoid) function



[Image source](#)

- Logistic function maps inputs to real interval $[0,1]$
 - **Can be *interpreted* as a probability distribution**

How do we Use Logistic Regression?

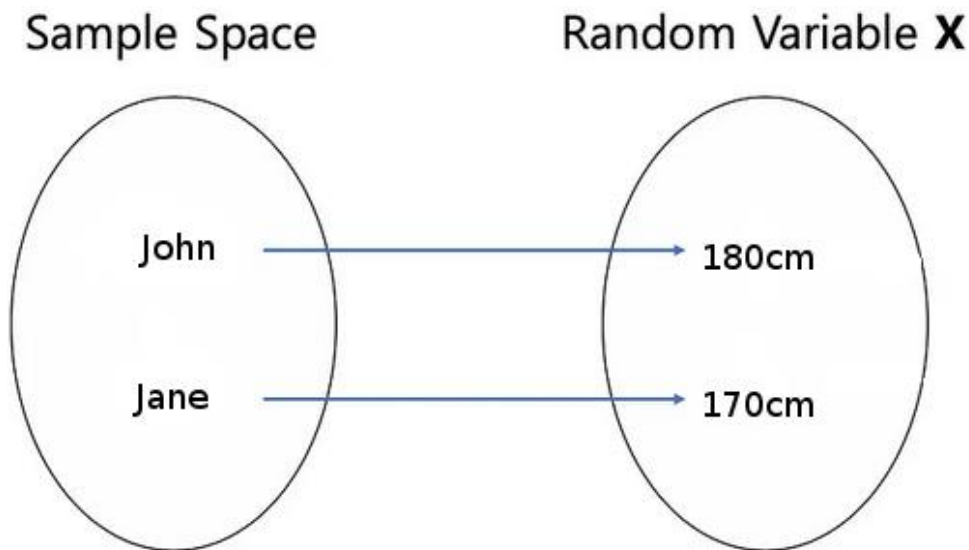
- **Binary** classification:
 - $P(y = 1) = \sigma(m x + b)$ --> interpreted as probability for positive class
 - $P(y = 0) = 1 - \sigma(m x + b)$ --> complementary probability for negative class
- That is, we get probabilities for both positive and negative class
 - We can do some *inference*, e.g. how much more likely is one class than the other?
 - But *prediction*?
- **Prediction:** we decide on a threshold
 - $p(y = 1) \geq 0.5$, we predict positive class
 - $p(y = 0) < 0.5$, we predict negative class
- We get the following **decision boundary**
 - $p(y = 1/x) = p(y = 0/x) = 0.5$
- Describing tasks and models with **probability theory** is very common!
 - We'll do this often, so let's review the basic concepts

Probability Theory Basics

- This is a **quick refresher** about **probability theory**
- This is later **useful for many things**
 - Describing tasks
 - Describing models, e.g. what language models do
 - More concepts commonly used in NLP research
- Today, some basic intuitions/definitions
 - Random variables
 - Probability distributions
 - Independence of events
 - Sum rule / product rule
 - Joint distributions
 - Conditional probability
 - Conditional independence
 - Bayes rule

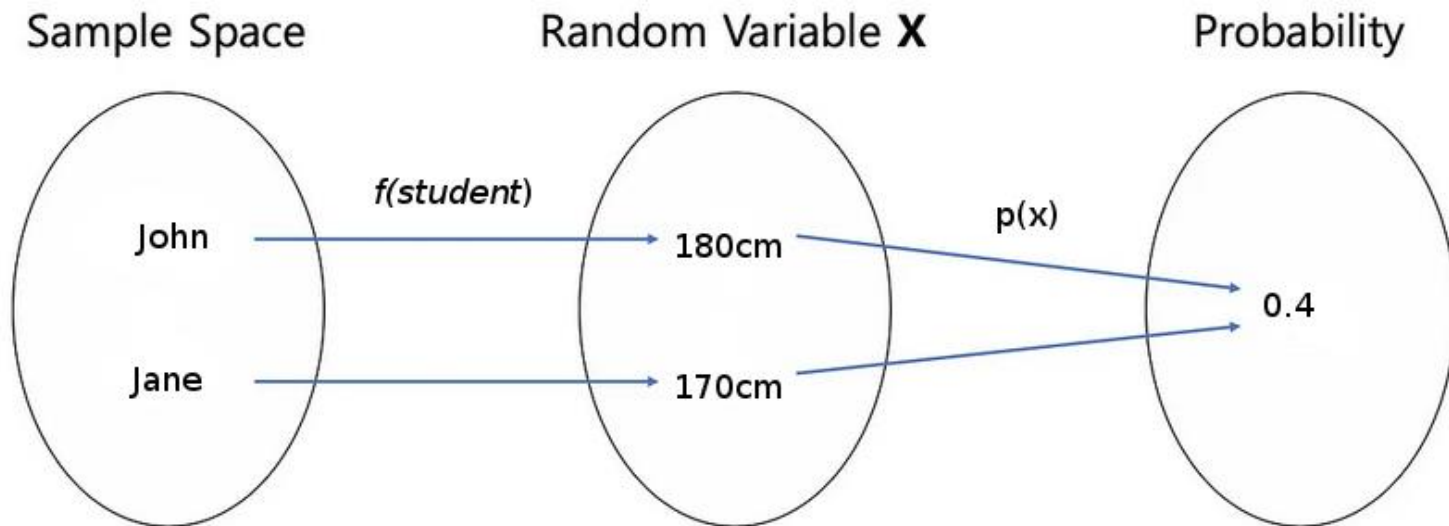
Random Variables

- **Function** between events and something relevant about the events
 - **Usually designed** according to something we want to model
- Example: say we are interested in average height of students
 - **Event:** randomly select a student
 - **Event space:** set of all students the course
- **Random variable X :** $f(\text{student}) = \text{height of student}$



Probability Distributions (1)

- Essentially a way to describe random variables
- **Random variables:** take values as a result of some random process
- **Distribution of random variable:** how often may each value be observed
- How does a distribution describe this?
 - A(nother) function $p(\mathbf{X})$: maps set of possible values a random variable can take to interval $[0,1]$



Probability Distributions (2)

- **Output of p for given event a :** probability of observing event a
 - Written as $p(X=a)$
 - Example: what is the probability of observing a height of 185cm?
 - $p(X = 185\text{cm}) = 0.1$, i.e. probability is 10%
- **Discrete distributions:** random variable can take finite set of values
 - p usually referred to as *probability mass function*
- **Continuous distributions:** r.v. can take infinite set of values
 - p usually referred to as *probability density function*
- Examples:
 - Bernoulli (discrete): models output of single binary experiment
 - Gaussian (continuous): normal distribution
- Is X in our students' height example discrete or continuous?
 - Depends on how to measure height, as reals or integers

Independence of events

- Events **A** and **B** are **independent** when one event happening does not affect the probability of the other happening
- For example:
 - Event **A**: Germany wins the world cup in 2014
 - Event **B**: Germany loses world cup semifinal
 - Event **C**: The stock market crashed in 2008
 - **Which of these events are dependent/independent?**
- A and B are not independent
 - If Germany loses the semifinal, it cannot win the final
- A and C are independent, as well as B and C
 - Very unlikely the 2008 crash has any impact on the 2014 world cup
- Independence perhaps best understood with **sum rule**

Sum Rule

- Probability of *union* of n events happening, e.g. **A OR B**
 - Formally: $p(A \cup B) = p(A) + p(B) - p(A \cap B)$
- For example: r.v. **Y** shows number in die, relevant events are:
 - **A**: “We observe a multiple of 3”, i.e. either 3 or 6
 - **B**: “We observe a multiple of 2”, i.e. either 2, 4 or 6
- Are **A** and **B** independent? **No**, see overlap in event spaces
- For $p(A \cup B)$ we subtract probability of overlapping events: $p(Y=6)$
 - $p(A) = 1/6 + 1/6 = 1/3$
 - $p(B) = 1/6 + 1/6 + 1/6 = 1/2$
 - $p(y = 6) = 1/6$
 - $p(A \cup B) = 1/3 + 1/2 - 1/6 = 0.66$
- With independent events, last term becomes zero:
 - Event **A**: “We observe an even number”, i.e. either 2, 4 or 6
 - Event **B**: “We observe an uneven number”, i.e. either 1, 3 or 5

Product Rule

- Probability of conjunction of n events happening, i.e. A **AND** B
- Formally: $p(A, B) = p(A/B)p(B)$
 - $p(A, B)$ = joint distribution of A and B
 - $p(A/B)$ = conditional distribution of A given B
- Side note:
 - All of these rules are *derived* from axioms of probability theory
 - For more, see probability theory literature
- Let's go over joint distributions and conditional distributions

Joint Distributions (1)

- Function $p(X_{1...n})$ from n random variables to interval $[0,1]$
 - Distribution of all possible combinations of those n random variables
- Easier to see with joint probability table
- For example, joint distribution of flipping two coins **A** and **B**:

A / B	Heads	Tails
Heads	0.1	0.2
Tails	0.3	0.4

- Note that all elements in table add up to 1, i.e. a distribution

Joint Distributions (2)

- We can derive many things from a joint distribution
- A very common operation: **marginalization**
 - Given joint distribution of A and B, derive distribution of A or B
- For example, we get distribution of coin A by summing over all possible values of B with the sum rule:

A/B	Heads	Tails	$p(A)$
Heads	0.1	0.2	= 0.3
Tails	0.3	0.4	= 0.7

- New column adds up to 1, i.e. a probability distribution
 - $p(A)$ known as **marginal distribution** of A
- This process is referred to as **marginalizing out B**
 - Normally expressed with sum rule: $p(A) = \sum_b p(A, B = b)$

Conditional Probability

- Probability of **seeing event A knowing event B was already seen**
- Formally, $p(A/B) = p(A,B)/p(B)$
- Intuitively, **space of all possible events shrinks** (for dependent RVs)
 - Imagine we toss a die
 - **A:** “We observe an even number”, i.e. either 2, 4 or 6
 - **B:** “We observe a multiple of 3”, i.e. either 3 or 6
 - After observing A, event space for B is not {3, 6}, it is {6}
- Note that $p(A/B)$ can be derived from joint distribution $p(A,B)$
 - Marginalize out A to get $p(B)$
 - Then you have everything you need

Conditional Independence

- Two events are **independent given a third event**
 - Observing **A** tells me something about **B**
 - But when observing **C**, **A** no longer says anything about **B**
- For example:
 - Event **A**: knowing a person has good vocabulary
 - Event **B**: knowing a person's height
- Are **A** and **B** independent?
 - No, since children have limited vocabulary
- Now consider event **C**: knowing a person's age
 - Given **C**, **A** and **B** are independent, since height implies age
 - Once we have age, height gives us no new information
 - Thus, **A** and **B** are conditionally independent given **C**

Bayes' Rule

- Apply product rule on conditional probability
- That is, $p(A/B) = p(A,B) / p(B) = p(B/A) p(A) / p(B)$
 - Note that we want probability of A given B, i.e. $p(A/B)$
 - And to get it, we use marginal distribution of A, i.e. $p(A)$
 - This may seem **counterintuitive**
- This is typically interpreted like so:
 - $p(A)$ known as **prior probability** distribution of event A
 - Describes **what we know/believe** about event A
 - $p(B/A)$ known as **likelihood**
 - Given event A, what is the likelihood that we get our data
 - $p(A/B)$ known as **posterior probability** of A
 - Interpreted as **updated description of A** after considering what we originally know about A (prior), as well as our data (event B)
- This interpretation gives rise to the *frequentists vs bayesian* debate
- And with this, we are **done with our probability refresher!**

A Probabilistic Perspective of ML

- **Task:** predict outcome of a coin tossed by your (unreliable) friend
 - Many factors: mass distribution, position in fingers, friction, etc.
 - Difficult to *model* physically
- Let's use a probabilistic model
 - Abstracts away complications of physical model
- **Probability:** uncertainty of some *random* event, e.g. coin toss
 - In practice: ***relative frequency of observations*** about the event
 - **Event in this case:** *seeing heads* when tossing the coin
- **Simple Model:** Bernoulli distribution
 - **Intuition:** it models outcome of *single binary* experiment
 - **Formally:** $p(k|\theta) = \theta^k (1-\theta)^{1-k}$ where k in $\{0,1\}$ is possible outcomes
 - **θ is the only parameter of model:** probability of seeing heads
- How do we get a value for this parameter so we can use our model?
 - We could **estimate this parameter**
 - We could blindly guess, or we could **use data for that estimation!**

Training Models: Parameter Estimation

- Say your friend allows you to observe ten tosses
 - You observe 3 out the 10 are heads
- You can estimate the value of Θ from this data
 - $\Theta = 3/10 = 0.3$ chance of seeing heads in a single toss
- We just counted **relative frequencies**
 - Training example: a single coin toss, i.e. y_i in $\{heads, tails\}$
 - Training data: 10 coin tosses
 - Relevant observations: we observe heads
- With a value estimated for parameter Θ , we can make predictions!
 - Our model has been trained and is ready to be used **on unseen data**
- Thus, **learning a function** means **estimating its parameters**
 - We don't know the parameters of a model
 - Instead, we have data and use it to estimate those parameters
- Let's look at a common framework for parameter estimation

Maximum Likelihood Estimation (1)

- Common approach for parameter estimation
 - Otherwise known as **MLE**
- Finds model that **maximizes likelihood** of observed data
 - But what does likelihood mean?
- Our model for the coin toss:
 - $p(k|\theta) = \theta^k (1-\theta)^{1-k}$ is a function of k with θ fixed (how we predict with it)
 - Let's **instead see it as** function of θ with fixed k , i.e. $L(\theta/k) = \theta^k (1-\theta)^{1-k}$
 - That is, a function of parameters given *fixed data* (assumed during training)
- We call **$L(\theta/k)$ the likelihood function**
 - It is **not a probability distribution**
 - Instead, we use it to find distributions that fit observed data
- How do we use the likelihood function during training?
 - We want distribution with *highest* (max) probability to our data
 - **Assumption:** set of observed data is most likely sample among all data

Maximum Likelihood Estimation (2)

- Formally, we want $\Theta = \max_{\Theta} L(\Theta/k)$
 - Equivalently, we want $\Theta = \operatorname{argmax}_{\Theta} \log p(k/\Theta)$
- First**, we make a common **assumption**: examples are **i.i.d.**
 - This stands for **independent and identically distributed**
 - Examples are independent, belong to same distribution
- Then**, we apply this assumption to our coin toss setting
 - Our model for a single example: $p(k/\Theta) = \Theta^k (1-\Theta)^{1-k}$
 - The **joint probability of N observed examples** is then given by:

$$\begin{aligned} P(D) &= P(y_1, y_2, \dots, y_N) \\ &= P(y_1)P(y_2) \dots P(y_N) \\ &= P(\text{head})^k P(\text{tail})^{N-k} \\ &= \theta^k (1 - \theta)^{N-k}. \end{aligned}$$

- Here we used the i.i.d. assumption from line 1 to line 2

Maximum Likelihood Estimation (3)

- **Finally**, to find this maximum value, we:
 - Compute the first derivative
 - Set it to zero
 - Extract a value for θ

$$\begin{aligned}\frac{\partial \log P(D)}{\partial \theta} &= \frac{\partial (\log \theta^k (1 - \theta)^{N-k})}{\partial \theta} \\ &= \frac{\partial (k \log \theta + (N - k) \log(1 - \theta))}{\partial \theta} \\ &= \frac{k}{\theta} - \frac{N - k}{1 - \theta} = 0 \\ \Rightarrow \hat{\theta} &= \frac{k}{N},\end{aligned}$$

- We obtained relative frequencies, as we had done already!
 - In practice, **MLE often does not have a closed-form solution**
 - Instead, **we rely on numerical methods**
- MLE related to many aspects of ML: cross entropy, MAP, etc.

Empirical Risk Minimization

- General training approach
- **Main goal:** estimate how a model will work in practice by **minimizing the average** of some **loss function** $L(x_i, y_i)$ on a training set
 - Here x_i is a training example and y_i its corresponding label
- Formally:

$$R_{\text{emp}}(h) = \frac{1}{n} \sum_{i=1}^n L(h(x_i), y_i).$$

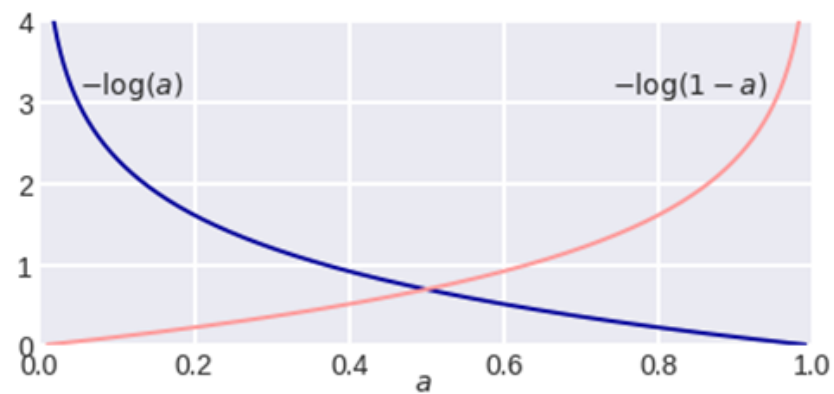
where n is size of training set.

- This is the most common training approach in ML/DL/NLP.
 - But what is a loss function, exactly?

What are Loss Functions?

- Functions that tell us **how a model performs during training**
 - Being a cost/loss, we want to **minimize it**
- Desiderata:
 - **Low loss:** when model prediction matches given label
 - The closer, the lower the loss/cost
 - **High loss:** when model prediction does not match label
 - The farther, the higher the loss/cost
- Example: **log loss**
 - Here a is model prediction
 - Note the desired behavior
 - Works with probabilistic models only

$$-\begin{cases} \log a_i, & y_i = 1, \\ \log(1 - a_i), & y_i = 0. \end{cases}$$



[Image source](#)

Let's Design a Training Objective

- Combine empirical risk minimization with log loss
 - Assume binary classification
- We have:
 - $J = - 1/N \sum_n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$
where N is size of training and p_i is prediction for example i
- Note the following:
 - Labels act as indicator factors
 - Log loss means bad model predictions get punished
 - We negate the entire expression so we can minimize it
- Thus, this is an optimization problem
 - This one can be shown to be a form of maximum likelihood estimation
- **Training objectives:** very important in NLP
 - E.g. masked vs autoregressive language have different, manually designed, training objectives

Summary: ML Refresher

- **Goal:** learn models that perform a task well on unseen data
- **Task:** from some input to some output
 - Regression: real-valued output
 - Classification: categorical output
- **Model:** abstract, often simplified version of a task
 - Comes with set of assumptions, e.g. linearity
- **Evaluation:** given predictions, compute relevant metric
- **Training:** parameter estimation
 - Objective functions (more not covered in *refresher*, e.g. regularization)
- All of these concepts are common in advanced NLP!
 - E.g. LLMs are **models** with trillions of **parameters**
 - Different LLMs **train** on different **objective functions** for different **tasks**
 - **Empirical risk minimization** and **MLE** used during training
 - They are **evaluated** on various **metrics** relevant to various tasks

Deep Learning Basics

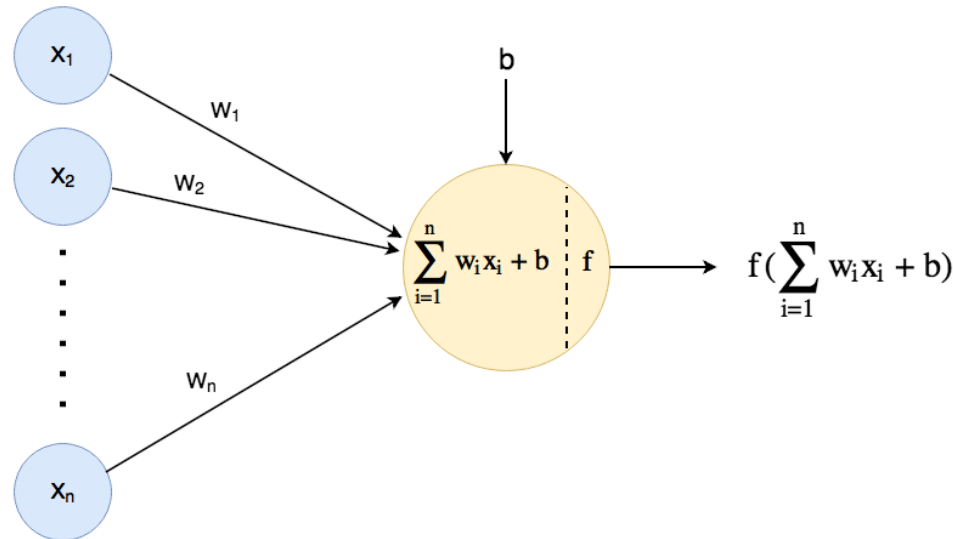
About this Section

- **Goal:** discuss basic concepts and terms that will be useful for the entire course
- **Outline:**
 1. Basic Deep Learning Concepts
 2. Backpropagation
 3. Language Models with Feed-forward Neural Networks

Mathematical Notation

- Scalars: a, b
- Vectors: $\mathbf{x}, \mathbf{y}$ (by default column vectors, i.e. size is $n \times 1$)
- Matrices: $\mathbf{X}, \mathbf{Y}$
- Scalar product: $a\mathbf{x}$
- Dot product: $\mathbf{x}^T \mathbf{w}$ (transpose because $\mathbf{x}$ is column vector)
- Matrix-vector product: $\mathbf{X}\mathbf{y}$
- Matrix-matrix product: $\mathbf{X}\mathbf{Y}$

What is an Artificial Neuron?

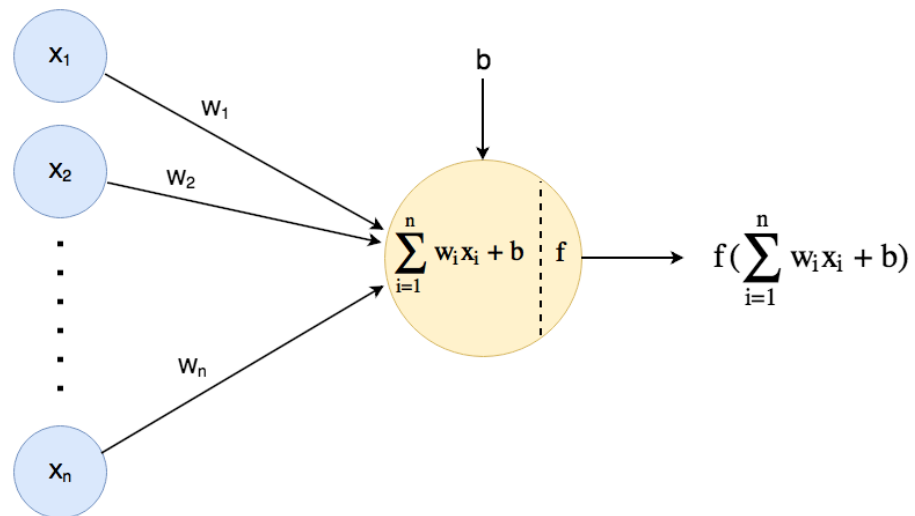


[Image source](#)

- Input is n -dimensional vector $\mathbf{x}$
- Features x_i multiplied with weights w_i , their products then added
 - What common operation is this?
 - A bias is optional, but almost always there, adds lots of flexibility to model
- Then an *activation function* f is applied to the result
- Different activation functions determine different artificial neurons

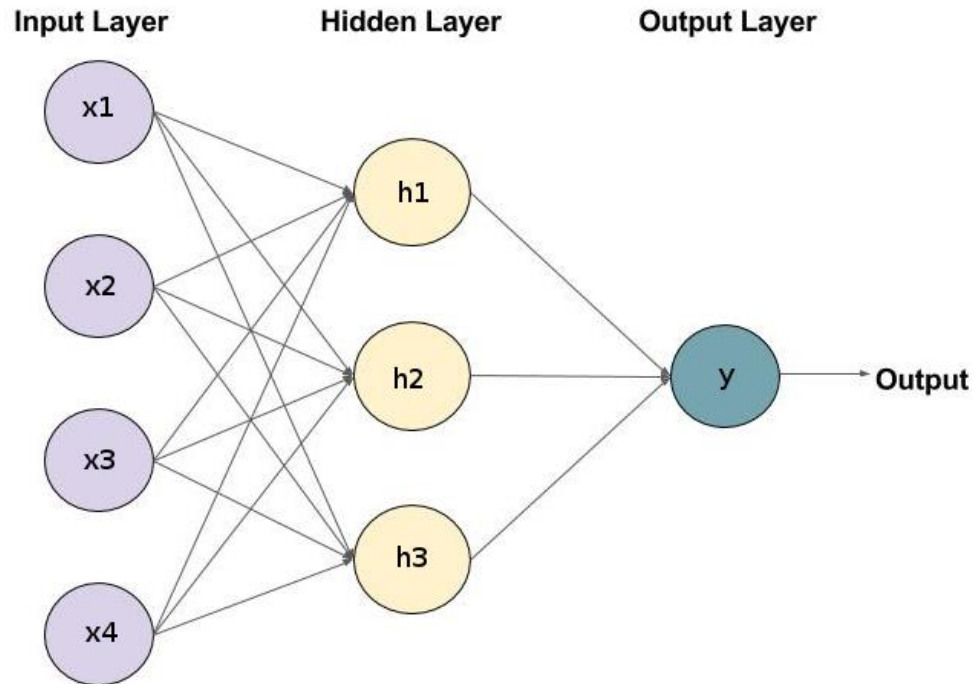
Types of Artificial Neurons

- They are mainly distinguished by their activation functions, e.g.
 - **Linear neuron:** $f(x) = x$
 - **Logistic neuron:** $f(x) = \sigma(x)$ (sigmoid function)
- What model do we obtain with a linear neuron?
 - Linear regression
- And with a logistic neuron?
 - Logistic regression
- More types:
 - **Tanh:** maps to $[-1,1]$
 - **ReLu:** $f(x) = \max(0,x)$
 - Etc.
- By adjusting weights:
 - We learn functions!



Feed-forward Neural Networks

- Let's do some **feature learning!**
 - We use *hidden* layers
 - Input to h_i : **x (our data)**
 - h_i : learned values
 - Input to y : **h (learned!)**
- If y is linear neuron:
 - Linear regression with learned features
- If y is logistic neuron:
 - Logistic regression with learned features
- Thus, FNNs are a framework for *designing* functions (*algebraic circuits*)
 - Information flows forward (but also backward during training, more later)
 - Main restriction: **no loops!** Computations seen as **directed acyclic graphs**



Operations in FNNs

- Example so far: **fully-connected neural network** (useful intuition)

- What does it compute?

- $\mathbf{x}$: $n \times 1$ input vector

- $\mathbf{h}$: $m \times 1$ hidden representation

- h_k has n w_{ij} , together: $\mathbf{w}_k$

- Let $\mathbf{W}_1$ be $n \times m$ matrix

- m columns, each a $\mathbf{w}_i$

- Then $\mathbf{h} = \mathbf{W}_1^T \mathbf{x}$ if h_k linear

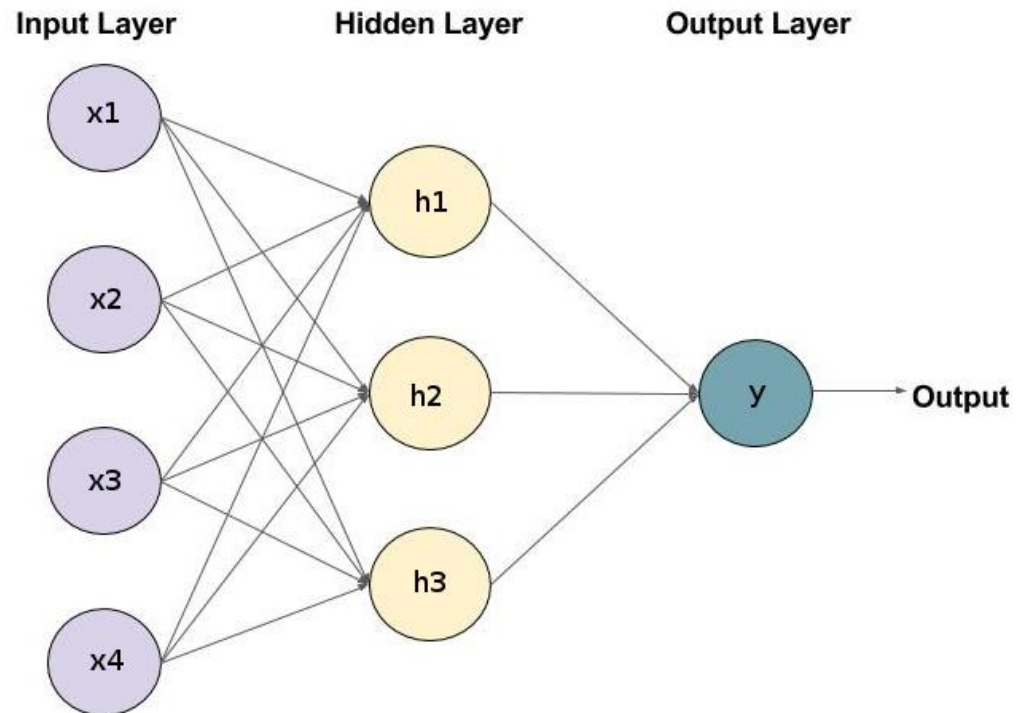
- $\mathbf{h} = f(\mathbf{W}_1^T \mathbf{x})$ w. activ. f

- $\mathbf{h} = f(\mathbf{W}_1^T \mathbf{x} + \mathbf{b})$ w. bias

- Similarly: $y = f(\mathbf{W}_2^T \mathbf{h} + b)$ where $\mathbf{W}_2$ is $m \times 1$, constructed as $\mathbf{W}_1$

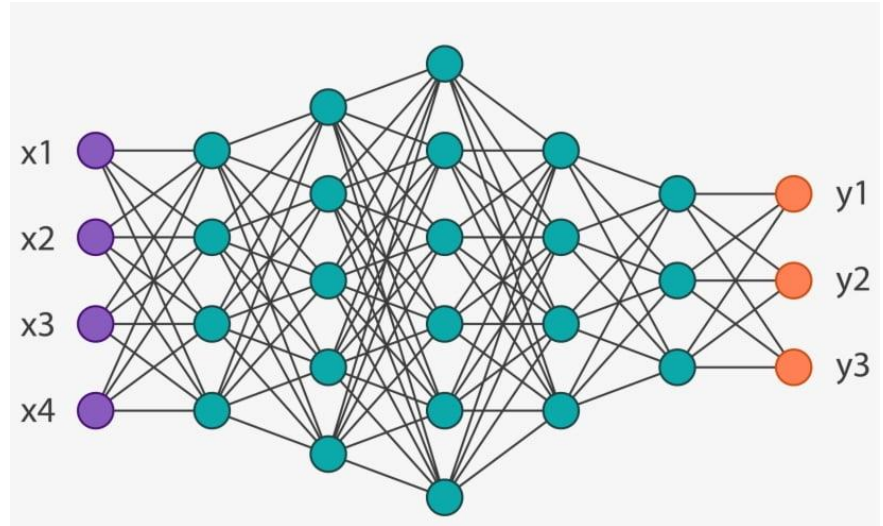
- Note network $y = \mathbf{W}_3^T \mathbf{x} + b$ exists with $\mathbf{W}_3 = \mathbf{W}_1 \mathbf{W}_2$, i.e. no hidden layers

- Why do we always use hidden layers then?



Why use Deep Networks?

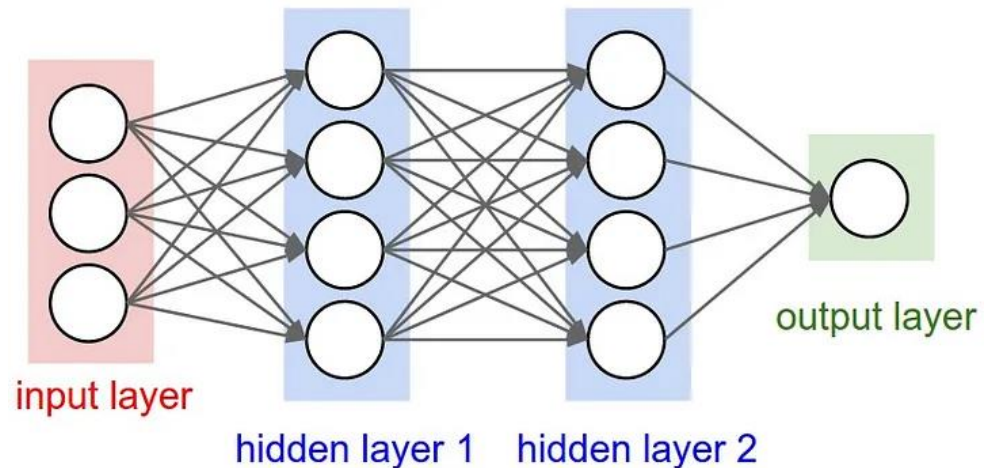
- When is a network deep?
 - Not well defined
 - Some say >4 hidden layers
 - Others say >2
- More important: **why?**
- Universality theorem
 - Formalities about FNN limits to approximate functions
 - One takeaway: no depth *needed*
- Why use depth then? **Hierarchical features!**
 - Each hidden layer is a learned feature vector, input vector to next layer
 - Component for designing networks: number and size of hidden layers
 - Smaller layers force models to learn compact but useful representations
 - More/larger hidden layers: more freedom, more parameters, difficult to train
- More parameters $\rightarrow$ stronger model, may overfit, requires lots of data



[Image source](#)

FNNs for Regression

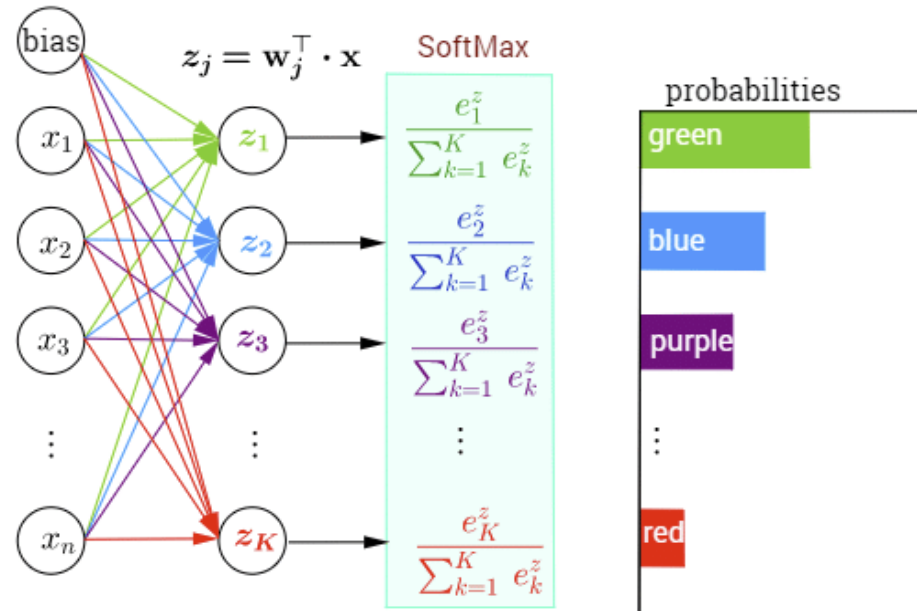
- Let's design a fully-connected network for regression
- **Input:** n -dimensional $\mathbf{x}$
- **Output:** real number y
- So, how many input units?
 - What type of units?
- How many output units?
 - What type of units?
- How many hidden layers?
 - What size/type of units?
- How to ensure this number of output units? **Linear layer!**
 - *Project* down to the size that you need
 - Remember: $y = \mathbf{W}_k^T \mathbf{h} + b$ where $\mathbf{W}_k$ is $m \times n$ matrix
 - $\mathbf{W}_k^T$ projects n dimensional vector to m dimensional vector space
- **Linear transformation** are **everywhere** in Deep Learning!



[Image source](#)

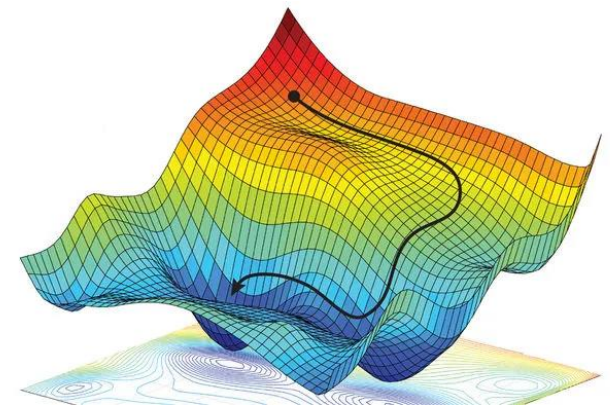
FNNs for Classification

- Let's design a fully-connected network for classification
- Input:** n -dimensional $\mathbf{x}$
- Output:** one of K classes
- So, how many input units?
 - What type of units?
- How many output units?
 - How do we ensure that?
 - What type of units?
- How to get predictions?
 - Softmax layer**
- Softmax function:** turns given vector into *probability vector* [Image source](#)
 - $\mathbf{x}$ is a probability vector if $x_i \in [0,1]$ for all i and $\sum_i x_i = 1$ (see first tutorial)
- Softmax layer:** linear layer for projection + softmax function
 - Input:** vector of logits; **Output:** probability vector for inference/prediction



Information Flow

- “Feed-forward neural networks: information flows forward”
 - Oversimplified! Important information flows backward *during training*
 - This influences design, inspired creation of important components
 - E.g. LSTMs, residual units, etc.
- Training often done with gradient-based optimizers
 - E.g. SGD, Adagrad, etc. (briefly discussed in first tutorial)
 - **Gradient information tells us how to update model parameters!**
- **Forward pass**
 - Compute function we are learning
- **Backward pass**
 - Compute gradient information to learn parameters
- How to compute gradients of functions computed by FNNs? **Backpropagation!**



[Image source](#)

Backpropagation (1)

- What is a derivative?
 - Let $y = f(h)$
 - dy/dh asks: **how much does y change when h changes?**
- A gradient is a multidimensional derivative
 - Let $\mathbf{h}$ be n -dimensional vector and $y = f(\mathbf{h})$ a function from R^n to R
 - $\partial y / \partial \mathbf{h} = (\partial y / \partial h_1, \partial y / \partial h_2, \dots, \partial y / \partial h_n)$ is gradient of y w.r.t. $\mathbf{x}$
 - I.e. the gradient of y w.r.t. $\mathbf{h}$ is also an n -dimensional (row) vector
 - Geometrically, **gradient vector is direction of the largest change of y**
- **What rate of change are we interested in during training?**
 - How much does loss L change when model parameters Θ change?
 - I.e. $\partial L / \partial \Theta$
- What if L is output of a compound parameterized function (FNN)?
 - E.g. $L = f(y)$ where $\mathbf{y} = g(\mathbf{h} / \Theta_2)$, $\mathbf{h} = z(\mathbf{x} / \Theta_1)$ (z is 1st layer, g is 2nd, f is loss)
 - Here $\partial \mathbf{y} / \partial \mathbf{h}$ is a Jacobian ($m \times n$ matrix where m is size of $\mathbf{y}$, n size of $\mathbf{h}$)
 - Same for $\partial \mathbf{h} / \partial \mathbf{x}$

Backpropagation (2)

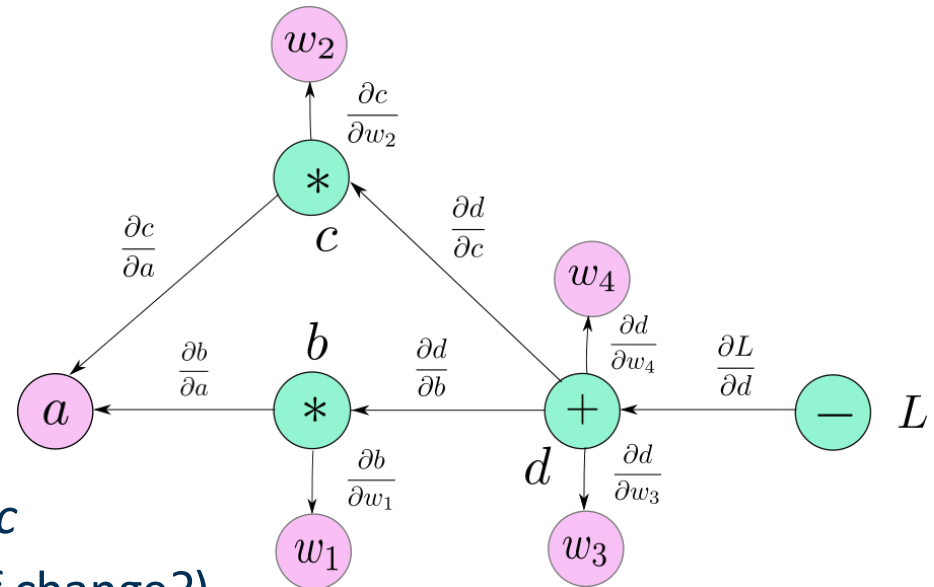
- We **propagate the change/error back** using the chain rule!

- Recall chain rule

- Let $y = g(h)$ and $h = f(\Theta)$
- $\partial y / \partial \Theta = \partial y / \partial h \cdot \partial h / \partial \Theta$

- Take this computation graph

- **Internal nodes:** operations
- **Leaves:** learnable weights w_i ,
input a



- Here $L = d - \text{label}$, $d = w_3b + w_4c$
- We want $\partial L / \partial w_4$ (which rate of change?)
- **Chain rule:** $\partial L / \partial w_4 = \partial L / \partial d \cdot \partial d / \partial w_4$
- Similarly, $\partial L / \partial w_2 = \partial L / \partial d \cdot \partial d / \partial c \cdot \partial c / \partial w_2$ where $c = w_2a$
 - Note that $\partial L / \partial d$ is useful for both $\partial L / \partial w_4$ and $\partial L / \partial w_2$
 - **Reusing gradients is common during backpropagation**

[Image source](#)

Backpropagation (3)

- Perhaps more important, take $\partial L / \partial w_2 = \partial L / \partial d \cdot \partial d / \partial c \cdot \partial c / \partial w_2$
 - Here, gradient $\partial c / \partial w_2$ is multiplied by factor $\partial L / \partial d \cdot \partial d / \partial c$
 - $\partial L / \partial d$ comes from *passing gradient* through minus operator in L
 - $\partial d / \partial c$ comes from *passing gradient* through plus operator in d
- Thus, **operations have impact on their "input" gradients!**
 - Important to consider when designing networks
 - E.g. *plus* operator has no effect on gradient
 - *Concat* operator splits gradient into two
- E.g. if either $\partial d / \partial c$ or $\partial L / \partial d$ are close to zero, $\partial L / \partial w_2$ will be small
 - In other words, we get little info about how to change w_2 to improve L
 - Generally, small gradients = slow/difficult to learn
- Many challenges in *training* deep networks, e.g. vanishing gradients
 - Inspired solutions like LSTMs, residual connections (used in transformers)
- **Let's look at a FNN for a practical NLP use case!**

Language Modeling Recap

- Predict the next word:
 - “Every Thursday there is a Schneckenhof ...”
- This suggests:
 - **Some words are more likely to appear than others given some context**
- Probabilistically:
 - $p(\text{meeting} / \text{“Every Thursday there is a Schneckenhof”})$
- **Generally: $p(w_{n+1} / w_1, w_2, \dots, w_n)$**
 - Conditional probability of w_{n+1} given joint distribution of $w_1, w_2, \dots, w_n$
 - w_i are words/tokens in some fixed vocabulary V the model can process
- **Such models can predict entire sequences with probability chain rule**
 - $p(w_1, w_2, \dots, w_n) = p(w_1)p(w_2/w_1)p(w_3/w_1, w_2) \dots p(w_n/w_1, w_2, \dots, w_{n-1})$
- Useful in many NLP settings, e.g. machine translation
 - A good language model should identify more common phrases
 - $p(\text{This sentence makes sense}) < p(\text{Makes sense this sentence does})$

Language Modeling with FNNs (1)

- [Bengio et al. \(2003\)](#) proposed FNN that simultaneously learns:
 1. “Distributed representations of words”
 2. “Probability function of word sequences expressed in terms of these representations”
- Size of word vectors (30, 60, 100) much smaller than vocabulary (17K)
- **Intuition: similar words have similar feature vectors**
- Thus, training sentence *The cat is walking in the bedroom* can be generalized to, e.g.:
 - *A dog is walking in a room*
 - *The cat is running in a room*
 - *A dog is walking in a bedroom*
 - *The dog is walking in the room*
- **How did they implement such a model?**
 - **A feed-forward neural network**
 - Not the first time it was done, but the first time at that scale

Language Modeling with FNNs (2)

- How did they model $p(w_m / w_1, w_2, \dots, w_{m-1})$?
 1. A mapping C between every word $w_i \in V$ to a m -dimensional vector $\mathbf{w}_i$
 Today known as an **embedding layer**.
 Given w_i , get row i in embedding matrix $\mathbf{C}$ (V rows, each a vector $\mathbf{w}_i$)
 1. Function g that maps input sequence of word vectors to a probability vector of size $|V|$, i -th component is probability of i -th word being next
- **Embeddings are parameters that are learned**
- Function g uses these embeddings plus additional parameters $\boldsymbol{\omega}$
 - $p(w_i / w_1, w_2, \dots, w_{m-1}) = g(i, \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{m-1} / \boldsymbol{\omega})$ where i is index of word w_i
- Overall, they learned function f , a composition of C and g , with a FNN
 - $f(i, w_1, w_2, \dots, w_{m-1}) = g(i, C(w_1), C(w_2), \dots, C(w_{m-1}) / \boldsymbol{\Theta})$
 - All model parameters are $\boldsymbol{\Theta} = (\mathbf{C}, \boldsymbol{\omega})$
- It's easy to see what C might look like, but what does g look like?
 - In other words, what did their network architecture look like?

Language Modeling with FNNs (3)

1st: Get word vectors w_i

2nd: Concatenate vectors to form x

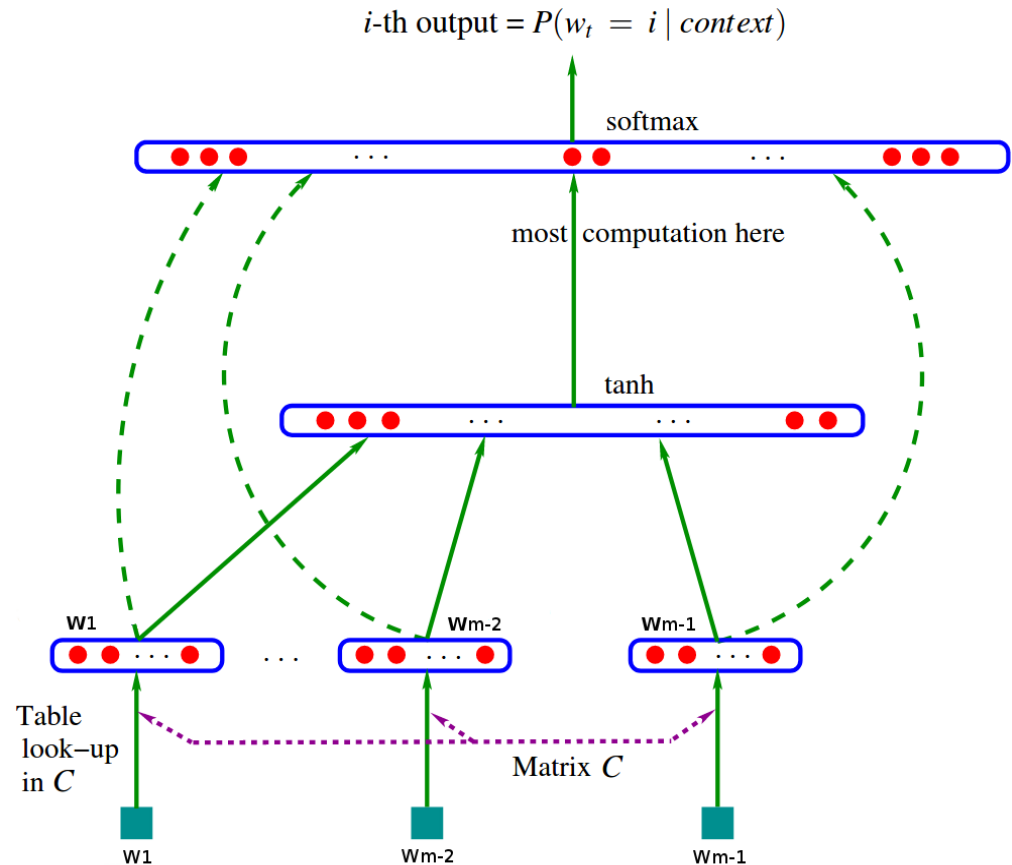
3rd: Hidden layer (matrix H transform. + $\tanh$ activ.)

4th: *Optional* linear layer w. matrix W from input to softmax layer (dashed lines)

5th: Softmax layer (matrix U transform. + softmax)

Overall: input to softmax

- $y = b + Wx + U \tanh(d + Hx)$ where b, d are biases
- Then $\Theta = (b, d, W, U, H, C)$



Language Modeling with FNNs (4)

- How did they train?
 - They estimated Θ that maximized likelihood of training data (i.e. **MLE**)
- Specifically:
 - $L = 1/T \sum_m \log f(w_1, w_2, \dots, w_{m-1}; \Theta) + R(\Theta)$
 - Here, T is number of words in training corpus
- Note a few things:
 - They average loss over the training set (empirical risk)
 - Penalizing term $R(\Theta)$ is regularization (weight decay on C, W, H, U)
- They trained in a **self-supervised** manner
 - I.e. yes, there are labels, but *no manual labeling* required
 - Instead, they used sequences of length m in the training corpus
 - For each sequence, first $m-1$ words are input, word m is target
- **Many of these components still in use today!**
 - Softmax layer, MLE, empirical risk, self-supervision

Summary: DL Basics

- **FNNs:** framework for constructing parameterized real-valued functions
 - We design and learn these functions (algebraic circuits)
- Information flows in both directions on these circuits
 - **Forward pass:** model expressivity
 - **Backward pass:** model “learnability”
- **Backpropagation:** training deep networks, chain rule
- **Important: we can learn useful features with deep models!**
 - Hidden layers interpreted as *learned representations*
 - Learned representations useful in other settings, i.e. transfer learning
 - E.g. style transfer, pre-training word representations
- **Language modeling with FNNs** by Bengio et al. (2003)
 - Established many modeling and training components still used today
- **Later in the course,** other types of FNNs relevant for NLP
 - **RNNs, transformers**

Word Embeddings

About this Section

- **Goal:** introduce concepts and methods about static word embeddings
- **Outline:**
 1. Sparse vs Dense Word Representations
 2. Word2Vec
 3. Properties of Word Vectors
 4. Evaluation of Word Vectors

What do we want to represent? (1)

- Words? I.e. strings of symbols such as *cat* or *dog*?
 - No, those are representations themselves!
 - What do they represent?
- **Meaning!**
 - We map strings such as *cat* or *dog* to concepts (or *senses*)
 - We refer to this mapping as *meaning*, what those words refer to
- **Is meaning a 1-to-1 mapping?**
 - **No**, words can have many meanings (remember ambiguity?)
- How do we, humans, handle word ambiguity? **Context!**
 - “I work at a bank.”
 - What’s more likely? A financial institution or by the river?
 - Again, context. Does the person work in Frankfurt? Are they fishermen?
- So, **we want to represent the relation between words and senses!**

What do we want to represent? (2)

- What do we want from such representations? Ideally usefulness, e.g.
 - Tell us if words have similar meanings, e.g. cat and dog
 - Tell us if words are antonyms, e.g. hot and cold
 - Tell us if they have positive or negative connotations, e.g. happy or sad
- **“Generally, a model of word meaning should allow us to draw inferences to address meaning-related tasks like question answering.”** Jurafsky and Martin, 2023
- Perhaps we should represent senses, then, not words!
 - Use something like WordNet, expert-crafted database of senses
- Databases like WordNet have not worked well in practice
 - Missing nuance, “good” is synonym of “proficient” only in some cases
 - New meanings missing, e.g. *genius*, *wicked*, *wizard*
- In practice, **dealing with word representations works better**
 - We focus on their relation to meaning (based on context)
 - We don’t commit to a given representation of sense

Sparse Word Representations

- Traditional approach: sparse word vectors
- Example: **one-hot vectors**
 - Given vocabulary V , represent each word with vector of size $|V|$ using one-hot encoding
- For example, given $V = \{cat, dog, airplane\}$
 - $V_{cat} = [1, 0, 0]$
 - $V_{dog} = [0, 1, 0]$
 - $V_{airplane} = [0, 0, 1]$
- **Problem 1:** $|V|$ can be very large
 - Recall Bengio's work from 2003: $|V| = 17K$
- **Problem 2:** No natural notion of similarity from such vectors
 - They are all equally different, e.g. with a dot product
 - But *cat* and *dog* are more similar to each other than to *airplane*
- Generally, **difficult to draw inferences from such representations**
 - Let's look at another approach that addresses these issues

Distributed Representations

- Idea from deep learning community (Geoffrey Hinton in 1986)
- Sparse representations:
 - Each representational component maps to single represented object
 - E.g. each feature in vector corresponds to one word
- **Distributed (or dense) representations:**
 - Many-to-many relation between representational components and represented objects
 - E.g. each word is represented with a set of (distributed) features
- **PROs:** model has freedom to use features “at will” during learning
- **CONs:** Less interpretable than, e.g. one-hot encoding
 - Given V , we know what word is represented by $[0, 1, 0]$
 - But what word is this? $[0.23, 0.447, 0.02]$
- Using **vectors to represent words** also has a **history in linguistics**
 - Osgood et al. (1957) represented words with 3-dimensional vectors
 - Each dimension encoded known useful (non-learned) features

Distributional Semantics (1)

- Idea from linguistics (Joos, 1950; Harris, 1954; Firth, 1957)
- Not the same concept!
 - *Distributed* representations != *distributional* semantics
 - In NLP, the latter can be seen as a special case of the former
- Let's guess what the word *foobar* means
 1. *Foobar* is often played in teams of 7.
 2. A *foobar* match is divided into 3 thirds of 25 minutes each, called runs.
 3. A *foobar* match is won by scoring 10 points.
- **What would you guess the word *foobar* refers to? Why?**
 - A type of food?
 - A sport?
 - A country?
- **Distributional hypothesis:**
 - “A word is characterized by the company that it keeps.” (Firth, 1957)

Distributional Semantics (2)

- Put differently:
 - We define the meaning of a word by its *distribution* in a language
 - I.e. by its neighboring words, grammatical environment
- **Words whose neighboring words are similar, have similar meanings.**
- We can use this idea to build useful **vector representations** of words!
- For example: **co-occurrence matrices**
 - Given defined *context*, e.g. a document, a sentence, a word window
 - Construct matrix C of size $V \times V$ where V is given vocabulary
 - Rows in matrix are words, columns are also words
 - m_{ij} denotes number of times word i appears in same *context* as word j (so, sparse)
- Co-occurrence matrices useful in many applications
 - E.g. latent semantic analysis (LSA) --> singular value decomposition of C
 - Extracts vector representations from (truncated) left-singular vectors
 - Similar count-based methods covered in basic Text Analytics course
- **Word embeddings: distributional hypothesis to learn dense vectors**

Word Embeddings

- **Idea:** to *embed* higher dimensional vectors into lower dimensional vector space such that (some of) the relative properties of the higher dimensional space are kept
 - Hence, **embedding** --> **(relatively) low-dimensional vector representation**
 - Low-dimensional compared to, e.g. $|V|$
 - Size typically between 50 and 1000 dimensions
 - Vectors are dense (distributed) representations
- Historically important approach: **word2vec** ([Mikolov et al. 2013](#))
 - Training based on distributional semantics
 - Done in the context of deep learning (though this model is not deep)
- Intuition:
 - **Instead of counting words for co-occurrence, let's train a classifier to predict whether two words are likely to appear in the same context.**
- Two approaches proposed for learning such a classifier
 - Skip-gram and continuous bag of words (CBOW) (we focus on [skip-gram](#))

The Skip-Gram Algorithm

- Algorithm for learning dense word vectors
- **Basic steps**
 1. Treat target words and neighboring context words as **positive examples**
 2. Randomly sample other words in vocabulary to get **negative examples**
 3. Train a classifier to predict whether two given words appear together
 4. Use learned weights as word embeddings
- **Why negative examples?**
 - In some tasks, negative examples often needed to avoid trivial solutions
 - E.g. push all vectors to same point, thus all vectors similar
 - If training objective is poorly designed, such a solution may "reach objective"
- Let's have a look at:
 1. **The task**
 2. **The model**
 3. **The training objective**

Skip-Gram: The Task

- Say we have the sentence:
 - ... *known as [ramen, a Japanese dish that] has recently become...*
- $c_1 \quad c_2 \quad w \quad c_3 \quad c_4$
- Here, w denotes **center word** and c_i denotes **context words**
 - I.e. context window is plus/minus a few words in this case
 - More context --> usually more useful, not always, more later
 - E.g. plus/minus 1 gives us *[a; ???; dish]*
 - Plus/minus 2 gives us *[ramen; a; ???; dish; that]* (assumes no punctuation)
 - **Task:** given tuple (w, c) , predict if c appears in context of w or not
 - So, binary classification
 - Probabilistically, $P(\text{True} | w, c)$
 - Similarly, $P(\text{False} | w, c) = 1 - P(\text{True} | w, c)$
 - How can we model these probabilities?

Skip-Gram: The Model (1)

- How to get probability $P(\text{True}/w,c)$?
 - **Key idea:** based on similarity of corresponding word embeddings
 - I.e. a word c is likely to occur near a target word w if their corresponding embeddings $\mathbf{e}_c$ and $\mathbf{e}_w$ are similar
 - Probabilistically, w and c are independent given their embeddings $\mathbf{e}_c, \mathbf{e}_w$
 - Let $\boldsymbol{\Theta} = [\mathbf{e}_1, \dots, \mathbf{e}_{|V|}]$. We have $P(\text{True}/w,c,\boldsymbol{\Theta})$ ($\boldsymbol{\Theta}$ often omitted for brevity)
- How can we represent that similarity?
 - Skip-gram went with **dot product**, i.e. $\mathbf{e}_c^T \mathbf{e}_w$
- But didn't we need a probability?
 - Ok, then $P(\text{True}/w,c) = \sigma(\mathbf{e}_c^T \mathbf{e}_w)$ where σ is the **sigmoid function**
 - Similarly, $P(\text{False}/w,c) = 1 - \sigma(\mathbf{e}_c^T \mathbf{e}_w) = \sigma(-\mathbf{e}_c^T \mathbf{e}_w)$
 - The last equality results from a property of the logistic function
- This is the probability of word c being in context of word w .
 - **What if I want to provide more context?** More context is better, right?

Skip-Gram: The Model (2)

- How do we compute $P(\text{True} | w, c_{1:L})$ where L is size of context window?
 - **Assumption: context words are independent of each other**
 - Computing joint distribution of words in context becomes much simpler
- Thus, we have

$$P(\text{True} | w, c_{1:L}) = \prod_{i \in L} \sigma(\mathbf{e}_i^T \mathbf{e}_w) = \sum_{i \in L} \log \sigma(\mathbf{e}_i^T \mathbf{e}_w).$$

- **How is this model parameterized?**
 - It actually learns *two* representations for each word
 - One for words as targets, one for words as context (just one vocabulary V)
 - Thus, parameterized by matrices $\mathbf{W}$ and $\mathbf{C}$, both of size $|V| \times d$ where d is a hyperparameter (embedding size)
 - *In other words, $\Theta = [\mathbf{W}, \mathbf{C}]$*
- How was this model trained?

Skip-Gram: The Training Objective (1)

- As with Bengio's LM from 2003: **self-supervision**
- Given text corpus (i.e. large sequence of text)
 - Iterate over sequences of size $n = L * 2 + 1$ (L is size of context window)
 - For each sequence, set center word as target, other words as context
- **For example:**
 1. ... *[known as ramen, a Japanese] dish that has recently become...*
 $C_1 \quad C_2 \quad W \quad C_3 \quad C_4$
 2. ... *known [as ramen, a Japanese dish] that has recently become...*
 $C_1 \quad C_2 \quad W \quad C_3 \quad C_4$
 3. ... *known as [ramen, a Japanese dish that] has recently become...*
 $C_1 \quad C_2 \quad W \quad C_3 \quad C_4$
- Thus, **for sequences 1, 2 and 3, you get positive examples (w, c) :**
 1. (ramen, known), (ramen, as), (ramen, a), (ramen, Japanese)
 2. (a, as), (a, ramen), (a, Japanese), (a, dish)
 3. (Japanese, ramen), (Japanese, a), (Japanese, dish), (Japanese, that)

Skip-Gram: The Training Objective (2)

- We also need **negative examples!**
- **Given w , we randomly sample words from V as candidates for c**
 - *For example:* (ramen, ostrich), (ramen, the), (ramen, Alaska), ...
 - They generated k (hyperparameter) negative examples per positive
- Then, given (w, c_{pos}) , corresponding (w, c_{neg_i}) , and cross entropy (CE), we have:

$$\begin{aligned} L_{CE} &= -\log \left[P(+|w, c_{pos}) \prod_{i=1}^k P(-|w, c_{neg_i}) \right] \\ &= -\left[\log P(+|w, c_{pos}) + \sum_{i=1}^k \log P(-|w, c_{neg_i}) \right] \\ &= -\left[\log P(+|w, c_{pos}) + \sum_{i=1}^k \log (1 - P(+|w, c_{neg_i})) \right] \\ &= -\left[\log \sigma(c_{pos} \cdot w) + \sum_{i=1}^k \log \sigma(-c_{neg_i} \cdot w) \right] \end{aligned}$$

- Generally, they applied empirical risk over all training examples
 - Since this is negative log likelihood (NLL), they did MLE of the training set

Task vs Goal

- **Task:** $P(\text{True} | w, c)$
 - What do we use that task for?
 - *Real-world uses?*
- **Goal:** learn word embeddings!
 - Task designed to force model to capture properties (distributional semantics)
 - Original task was different, $P(w_{t+j} | w_t)$
- Other models have tasks and goals
 - E.g. GPT-style LLMs are autoregressive language models
 - BERT-style models are masked language models
- **Sometimes we want to use the model to make predictions**
 - **Other times, we just want their learned weights**

Other Types of Static Embeddings

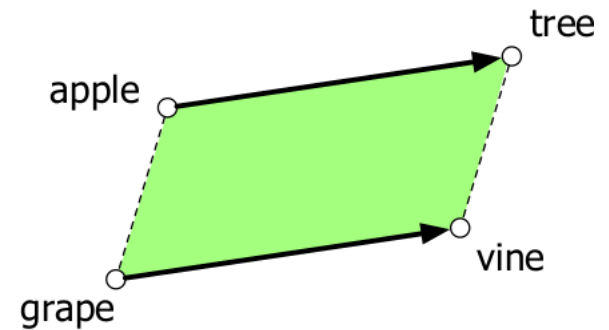
- **Word2vec is useful model for intuition/learning about embeddings**
 - It was also successful and is historically important
 - But there are other ways of learning word embeddings
 - These are often used in practice as well
- **GloVe:** stands for Global Vectors (Pennington et al., 2014)
 - Combines count-based models with methods like word2vec
 - Constructs co-occurrence matrix
 - Learns matrix factorization with least squares objective
 - Covered in basic Text Analytics course
- **fasttext:** addresses problem of unknown words in word2vec
 - That is, new words in test that were not present in training
 - For this, they used representations based on subwords
 - Also covered in basic Text Analytics course
- Static embeddings have some **nice and useful properties!**
 - At the time, they were quite impressive!

Properties of Word Embeddings (1)

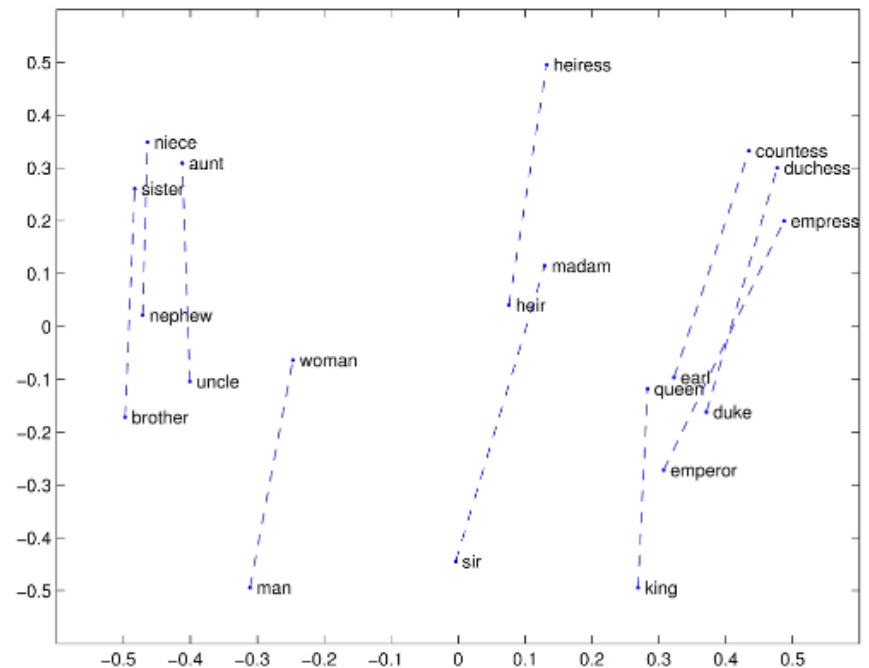
- **Size of context window is important!** (Levy and Goldberg, 2014)
 - It affects the representations we learn!
- **Smaller sizes tend to provide more static embeddings**
 - Depends on immediate neighborhood
 - Similarity tends to be based on same parts of speech
- E.g. what is similar to Hogwarts with small context window?
 - Other fictional schools, e.g. Sunnydale (from Buffy the Vampire Slayer)
- **Larger sizes tend to focus on similarity based on topic**
 - So words with same part of speech not necessarily related
- E.g. what is similar to Hogwarts with larger context window?
 - Dumbledore, Malfoy, half-blood
- **This was observed for skip-gram**
 - Different methods may differ in such properties
 - But it is safe to assume the context window has an impact

Properties of Word Embeddings (2)

- They capture relational similarity
 - Parallelogram model from cognitive science



- Example from GloVe
 - Skip-gram known to capture this
 - It's not as smooth all the time



Evaluating Word Embeddings

- **Intrinsic evaluation:**
 - Correlation between similarities of representations with expert-given similarity scores
- Examples:
 - WordSim353 (2002): 353 noun pairs, e.g. (*plane, car*)
 - SimLex-999 (2015): more difficult, e.g. (*cup, mug*)
- Another approach: **analogy task** (e.g. *man + woman – king = queen*)
 - Word2vec used this in their original work
- **Most important evaluation: extrinsic!**
 - I.e. using the representations in NLP *downstream tasks* and see if performance improves

Summary: Word Embeddings

- **Vector representations** of words
- **Sparse vs dense** (distributed) representations
 - PROs and CONs to each
 - Different methods to learn them (each a different task)
- **Distributional semantics**
 - Important principle
 - Commonly used to learn word embeddings, language models
 - Spoiler: also used when training LLMs
- Classic way to learn dense word representations: **word2vec**
 - Specifically, the skip-gram task, model and training objective
 - Resulting learned representations have **nice properties**
- **In the future**, other tasks for other goals
 - E.g. masked and autoregressive language models

References

- Natural Language Processing: A Machine Learning Perspective, Zhang et al, 2018
 - Chapters 2, 3, 4 and 13
- Speech and Language Processing, Jurafsky et al., 2023
 - Chapter 3, 6
- Natural Language Processing, Eisenstein, 2018
 - Chapter 14